

1 Introduction

Diabetic retinopathy (DR) is a microvascular complication of diabetes mellitus and one of the leading causes of preventable vision loss in working-age adults. Pooled population studies estimate that roughly one in three people with diabetes has some degree of DR and about one in ten has a vision-threatening form [1]; a more recent meta-analysis puts the global prevalence among people with diabetes at about 22% and projects that the absolute number of affected people will keep rising through 2045 [2]. Because early DR is asymptomatic and timely laser or anti-VEGF treatment prevents most severe vision loss, international guidelines recommend regular screening of every person with diabetes [3].

Screening is usually performed with colour fundus photography. A trained grader assigns each image to one of the five levels of the International Clinical Diabetic Retinopathy (ICDR) severity scale [4]: no apparent retinopathy (grade 0), mild non-proliferative DR with microaneurysms only (grade 1), moderate non-proliferative DR (grade 2), severe non-proliferative DR (grade 3) and proliferative DR (grade 4). Grades 2 and above are generally considered *referable* and require specialist review. Manual grading is accurate when done by experienced readers, but it is slow, requires a scarce workforce, and shows non-negligible inter-grader variability [5]. In many low- and middle-income regions, where the diabetic population is growing fastest, the number of trained graders is far below what population screening would require [3].

Deep learning has changed what is technically possible. Convolutional neural networks trained on tens of thousands of graded photographs reach sensitivity and specificity comparable to ophthalmologists for referable DR [6–8], systems trained on very large cohorts cover the whole disease spectrum [9], and an autonomous AI system has been evaluated in a prospective pivotal trial in primary care [10]. A systematic review found the diagnostic performance of deep learning to be broadly equivalent to that of health-care professionals, while warning that few studies used external validation or reported their methods completely [11].

Why grading on small datasets remains hard

These successes rely on large, often private datasets. Many research groups and most new clinical sites have access only to a few hundred labelled images, such as the Indian Diabetic Retinopathy Image Dataset (IDRiD) [12, 13]. In this regime three properties of the task become limiting.

(i) Small, sparse evidence. Adjacent ICDR grades are separated by small findings. A single microaneurysm, a few pixels wide in a downsampled image, moves an image from grade 0 to grade 1; the number and distribution of haemorrhages and venous beading separate grade 2 from grade 3; and neovascularisation defines grade 4. A network that summarises the whole image by global average pooling gives every location the same weight, so a handful of lesion pixels can be diluted by the large area of healthy retina [14, 15].

(ii) Ordered labels. The grades are ordinal. Predicting grade 4 for a healthy eye is clinically far worse than predicting grade 1, but standard cross-entropy penalises both errors equally. The quadratic weighted kappa (QWK) used by most benchmarks [16] reflects this ordering, and so should the model and its decision rule [17–19].

(iii) Imbalance and instability. Mild and severe grades are under-represented, and with few images per class both the learned weights and any post-hoc decision thresholds are noisy. Small datasets also make it tempting, and easy, to tune choices on the test set, which inflates reported performance [20, 21].

Our approach

We address these three issues jointly in a compact model, NITSCI, and in an evaluation protocol designed for small datasets. For (i), the backbone features are processed by two experts: a *global expert* that relates all retinal regions through self-attention, and a *lesion expert* that sees the same features re-weighted by a learned lesion-attention map. Rather than fusing them by

simple concatenation, the experts exchange information through bidirectional cross-attention, so that lesion evidence is interpreted in its anatomical context and global context is focused on the lesions. An image-adaptive gate, in the spirit of mixtures of experts [22, 23], decides how much each expert contributes to the final representation. For (ii), a classification head and an ordinal regression head are trained together and combined into one continuous grade score. For (iii), the model is first trained on the larger APTOS-2019 dataset [24] and then fine-tuned on IDRiD with five-fold cross-validation; class-balanced loss weights, weight averaging and a fold ensemble with test-time augmentation stabilise the predictions; and every data-dependent choice, including the decision thresholds, is made on out-of-fold predictions while a held-out test set is used exactly once.

Contributions. The main contributions of this work are:

1. **A cross-expert lesion-aware head.** We propose a head that splits backbone tokens into a global and a lesion stream, couples them through bidirectional cross-attention, and fuses them with an image-adaptive gate. The lesion stream is learned from grade labels alone, without lesion annotations. The head adds 1.40 M parameters to a 19.85 M-parameter EfficientNetV2-S backbone.
2. **An ordinal dual-head decision rule.** The expected grade of the classification head and the output of a regression head are averaged into a continuous score whose four cut-points are fitted on out-of-fold data.
3. **A leakage-free protocol for small datasets.** Cross-dataset pretraining, stratified five-fold fine-tuning, an ensemble with test-time augmentation, and a held-out test set that is never used for training, model selection or threshold fitting; results are reported with bootstrap confidence intervals and paired McNemar tests.
4. **Component-level evidence.** Seven controlled ablations, run with identical splits and seeds, isolate the effect of the lesion map, cross-attention, gate, ordinal head, pretraining and pre-processing on both five-class grading and referable-DR screening.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the datasets, data partitioning and pre-processing. Section 4 presents the NITSCI architecture, training objective and decision rule. Section 5 gives implementation details, metrics and statistical methods. Section 6 reports the results and ablations, Section 7 discusses their implications and limitations, and Section 8 concludes.